25901443 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY204 - MATERIAL CHARACTERISATION TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. Write Bragg's law and explain the terms. [K] / [CO1]
2. Name any two commonly used laser sources in Raman spectroscopy. [U] / [CO2]
3. Explain how conjugation affects the position of absorption maxima in UV–Visible spectrum [U] / [CO2]
4. How are liquid samples prepared for infrared (IR) spectroscopy? [U] / [CO2]
5. Why is the lifetime of phosphorescence greater than fluorescence? [K] / [CO2]
6. What are the applications of Scanning Electron Microscopy (SEM) in materials science? [U] / [CO3]
7. Name the source used for generating electrons in an electron microscope. [U] / [CO3]
8. Why do electron microscopes achieve higher resolution than optical microscopes? [U] / [CO3]

9. What information is obtained in the spectroscopic mode of Scanning Tunneling Microscopy? [U] / [CO3]

10. Explain phonons in the context of sample - electron beam interaction. [U] / [CO3]

11. What is the role of a temperature sensor in thermal analysis technique? [U] / [CO4]

12. Sketch the thermogravimetric curve for multistage decomposition with and without a stable intermediate. [K] / [CO4]

13. Distinguish between Cahn microbalance and null point microbalance used in thermogravimetry. [U] / [CO4]

14. Give a short note on principle of Differential Thermal Analysis. [U] / [CO4]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. How can crystallite size be calculated from XRD peak broadening? [U] / [CO1]

16. Describe the mechanism of Raman scattering in detail. [U] / [CO2]

17. Describe the main components of a UV spectrophotometer. [K] / [CO2]

18. What insights can we gain from qualitative analysis of EDS data? Explain with a graph. [U] / [CO3]

19. Summarize how capillary forces affect tip-sample interactionsin an Atomic force microscope. [U] / [CO3]

20. Explain the functions of probe and scanner in Scanning Probe Microscopy. [U] / [CO3]

21. Distinguish between conventional Differential Scanning Calorimetry and temperature-modulated Differential Scanning Calorimetry. [A] / [CO4]

22. Explain the solid phase transformation and thermal decomposition.Give example. [U] / [CO4]

[5x6 = 30]